7.11 PreQuiz: Logarithmic and Exponential Functions — Markscheme

1. [Maximum mark: 6]

   Maya invests \$1800 at 5 % per annum, compounded annually.

   (a) End-of-year-1 value $= 1800 \times 1.05 = 1890$. **AG**

   (b) $V_8 = 1800(1.05)^8$ **M1**

   $V_8 = \$2659.42$ **A1**

   (c) Need the smallest integer n with $1800(1.05)^n > 2700$. **M1**

   $n > \dfrac{\ln(2700/1800)}{\ln(1.05)} = 8.31\ldots$ **A1**

   Smallest full-year $n = 9$ years. **A1**

   Alternative method (table): show both adjacent rows $1800(1.05)^8 = 2659.42$ *and* $1800(1.05)^9 = 2792.39$*, then state* $n = 9$*. A single-row table earns full marks but should receive a feedback note recommending the adjacent row to justify the crossover.*

2. [Maximum mark: 6]

$F(t) = 1200e^{0.18t}$.

(a) $F(1) = 1200e^{0.18} = 1436.66\ldots$, so about 1437 followers. **A1**

(b) $F(4) = 1200e^{0.72}$ **M1**

$F(4) = 2465.32\ldots$, so 2465 followers. **A1**

(c) Set $1200e^{0.18t} = 3000$. **M1**

$e^{0.18t} = 2.5$, so $0.18t = \ln 2.5$. **M1**

$t = \dfrac{\ln 2.5}{0.18} = 5.09$ years. **A1**